Math 231A HW 4

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October 30, 2025

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Problem 1

Etingof Problem Set 3.13:

(1) Let $\mathfrak{g}$ be a three-dimensional real Lie algebra with basis x, y, z and commutation relation $[x, y] = z$, $[z, x] = [z, y] = 0$ (this algebra is called *Heienberg Algebra*). Without using Campbell-Hausdorff formula, show that in the correspondin Lie group, one has $\exp(tx)\exp(sy) = \exp(tsz)\exp(sy)\exp(tx)$ and construct explicitly the connected, simply-connected Lie group corresponding to $\mathfrak{g}$.

(2) Generalize the previous part to the Lie algebra $\mathfrak{g} = V \oplus \mathbb{R}z$, where V is a real vector space with non-degenerate skew-symmetric form w and the commutation relations are given by $[v_1, v_2] = w(v_1, v_2)z$, $[z, v] = 0$.

Solution: Since both questions have nearly the same method, so we'll state and prove the generalized case in (2), then go back and prove (1).

(2)

Lemma: Part (2)

Given the Lie algebra $\mathfrak{g} = V \oplus \mathbb{R}z$, where V is a finite-dimensional real vector space with non-degenerate skew-symmetric form w.

Define the Lie Bracket $[\cdot,\cdot] : \mathfrak{g}\times\mathfrak{g} \to \mathfrak{g}$ by commutation relation $[v_1, v_2] = w(v_1, v_2)z$ and $[z, v] = 0$ for all $v, v_1, v_2 \in V$. Let G be the simply-connected Lie Group corresponding to $\mathfrak{g}$, then the following formula always holds,for all $s, t \in \mathbb{R}$ and $v_1, v_2 \in V$:

$$\exp(tv_1)\exp(sv_2)\exp(-tv_1)\exp(-sv_2) = \exp(ts\cdot w(v_1, v_2)z) \tag{1.1}$$

Proof: To prove the above statement, we'll approach using the following:

I. Center of $\mathfrak{g}$ is $\mathbb{R}z$:

Given that $\mathfrak{g} = V \oplus \mathbb{R}z$, then every $v \in \mathfrak{g}$ can be uniquely expressed as $v_1 + kz$, where $v_1 \in V$ and $k \in \mathbb{R}$. Which, given that $u = v_2 + k'z$ (with $v_2 \in V$ and $k' \in \mathbb{R}$) with $u \in \mathfrak{z} :=$ center of $\mathfrak{g}$, then we must have $[u, v] = 0$ for all $v \in \mathfrak{g}$. Hence, the following is true:

$$0 = [u, v] = [v_2 + k'z, v_1 + kz] = [v_2, v_1] + k[v_2, z] + k'[z, v_1] + kk'[z, z] = w(v_2, v_1) \tag{1.2}$$

This implies that $w(v_2, v_1) = 0$ for every $v_2 \in V$. However, with w being a non-degenerate bilinear form, if $v_2 \neq 0$, one must have some corresponding $v_{2'} \in V$ such that $w(v_2, v_{2'}) \neq 0$. Hence, this enforces $v_2 = 0$, so $u = k'z \in \mathbb{R}z$. This shows that $\mathfrak{z} \subseteq \mathbb{R}z$.

Also, notice that given any $k \in \mathbb{R}$, any $v = v_1 + k'z \in \mathfrak{g}$ (where $v_1 \in V$ and $k' \in \mathbb{R}$) satisfies the following:

$$[kz, v_1 + k'z] = k[z, v_1] + kk'[z, z] = 0 \tag{1.3}$$

Hence, we also have $kz \in \mathfrak{z}$, or $\mathbb{R}z \subseteq \mathfrak{z}$. Therefore, $\mathfrak{z} = \mathbb{R}z$, this is the center of the Lie algebra $\mathfrak{g}$.

Moreover, this also implies that every $t \in \mathbb{R}$ satisfies $\exp(tz) \in Z(G)$ (since $\mathfrak{z}$ the center of $\mathfrak{g}$, is the tangent space of $Z(G)$).

II. Expression of Group Commutator:

Recall that given any $v_1, v_2 \in V$ and $t, s \in \mathbb{R}$, we have $\exp(tv_1)\exp(sv_2)\exp(-tv_1) = \mathrm{Ad}(\exp(tv_1)) \cdot \exp(sv_2)$, and such expression is the same as $\exp\big(\mathrm{Ad}_{\exp(tv_1)}(sv_2)\big)$ (where $\mathrm{Ad}_{\exp(tv_1)} \in \mathrm{GL}(\mathfrak{g})$ denotes the differential of $\mathrm{Ad}(\exp(tv_1))$ at $1 \in G$). However, remember that for $\mathrm{ad}(tv_1) \in \mathrm{End}(\mathfrak{g})$, it satisfies $\exp(\mathrm{ad}(tv_1)) := \sum_{n=0}^{\infty} \frac{(\mathrm{ad}\ (tx))^n}{n!} = \mathrm{Ad}_{\exp(tv_1)}$ as linear operator on $\mathfrak{g}$. So, we get the following:

$$\begin{aligned}
\mathrm{Ad}_{\exp(tv_1)}(sv_2) &= \sum_{n=0}^{\infty} \frac{1}{n!}(\mathrm{ad}(tv_1))^n \cdot (sv_2) \\
&= sv_2 + \mathrm{ad}(tv_1)(sv_2) + \sum_{n=2}^{\infty} \frac{1}{n!}(\mathrm{ad}(tv_1))^{n-2}\Big((\mathrm{ad}(tv_1))^2 \cdot (sv_2)\Big) \\
&= sv_2 + [tv_1, sv_2] + \sum_{n=2}^{\infty} (\mathrm{ad}(tv_1))^{n-2} \cdot [tv_1, [tv_1, sv_2]] \\
&= sy + ts \cdot w(v_1, v_2)z + \sum_{n=2}^{\infty} (\mathrm{ad}(tv_1))^{n-2} \cdot [tv_1, ts \cdot w(v_1, v_2)z] \\
&= sy + ts \cdot w(v_1, v_2)z
\end{aligned} \tag{1.4}$$

(Note: recall that $[v_1, z] = 0$ for all $v_1 \in V$).

Hence, we get the formula $\exp(tv_1)\exp(sv_2)\exp(-tv_1) = \exp\big(\mathrm{Ad}_{\exp(tv_1)} \cdot (sy)\big) = \exp(sy + ts \cdot w(v_1, v_2)z)$. And, by the fact that $\exp(kz)$ is central for all $k \in \mathbb{R}$ (the claim in **I.**), we have $\exp(tv_1)\exp(sv_2)\exp(-tv_1) = \exp(sy + ts \cdot w(v_1, v_2)z) = \exp(sy)\exp(ts \cdot w(v_1, v_2)z)$. Hence, we end with the following:

$$\begin{aligned}
\exp(tx)\exp(sy)\exp(-tx)\exp(-sy) &= \exp(sy)\exp(ts \cdot w(v_1, v_2)z)\exp(-sy) \\
&= \exp(sy)\exp(-sy)\exp(ts \cdot w(v_1, v_2)z) \\
&= \exp(ts \cdot w(v_1, v_2)z)
\end{aligned} \tag{1.5}$$

Take $t, s = 1$, we recover what the lemma wants. $\square$

For the actual Lie Algebra and Lie Group of that, consider the Generalized Heisenberg Algebra:

For this, we do need to show that $k = \dim(V) = 2n$ for some $n \in \mathbb{N}$ first though. Since w is a non-degenerate skew-symmetric bilinear form on V, then pick any nonzero $v_1 \in V$, the linear functional $w(v_1, \_) : V \to \mathbb{R}$ is nontrivial (so $\ker(w(v_1, \_)) = k - 1$), hence there exists nonzero $u_1 \in V$, such that $w(v_1, u_1) \neq 0$ (in particular, can choose u_1 such that $w(v_1, u_1) =$

1). Also, notice that because w is skew-symmetric, then $u_2 \notin \text{span}\{v_1\}$ (if so, then the output should be 0). So, the two are in fact linearly independent.

Now, consider $V_1 = \ker(w(v_1, \_)) \cap \ker(w(u_1, \_))$: If such set is 0 then we're done (where $n = 1$), but if not, then notice that since $V = \text{span}\{v_1\} \oplus \ker(w(u_1, \_)) = \text{span}\{u_1\} \oplus \ker(w(v_1, \_))$ (since $v_1 \notin \ker(w(u_1, \_))$ and $u_1 \notin \ker(w(v_1, \_))$, while the dimension works out), then based on the claim of direct sum (together with the fact that v_1, u_1 are linearly independent), every vector $v \in V$ has $v = av_1 + bu_1 + v'$ for unique $a, b \in \mathbb{R}$ and v' in the intersection of the kernel. Hence, $V = \text{span}\{v_1\} \oplus \text{span}\{u_1\} \oplus V_1$, showing that $\dim(V_1) = k - 2$.

Then, notice that now w restricts to another non-degenerate skew-symmetric bilinear form on V_2: For every nonzero $v \in V_1$, if u satisfies $w(v, u) \neq 0$, $u \notin \text{span}\{v_1, u_1\}$, since we have $w(v_1, v) = w(u_1, v) = 0$. So, it implies $u = av_1 + bu_1 + v'$ for some $a, b \in \mathbb{R}$ and $v' \in V_1$, where $w(v, v') = w(v, av_1 + bu_1 + v') = w(v, u) \neq 0$, hence one can find $v' \in V_1$ such that $w(v, v') \neq 0$. Hence, since $\dim(V_1) = k - 2 < k$, apply induction on the claim, we get that $\dim(V_1)$ is even, showing that $k = \dim(V_1) + 2$ is again even.

Now, consider the following subspace of $\mathfrak{gl}_{2n+2}(\mathbb{R})$, defined as $\mathfrak{h}$:

$$\mathfrak{h} := \text{span}\{v_1, ..., v_n, u_1, ..., u_n\} \oplus \mathbb{R}z \tag{1.6}$$

$$\text{where} \quad v_i = \begin{pmatrix} 0 & \dots & 1 & \dots & 0 \\ \vdots & \ddots & \vdots & & \vdots \\ 0 & \dots & 0 & \dots & 0 \\ \vdots & & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix} \text{at the} \quad (i+1)^{\text{th}} \text{ column} \tag{1.7}$$

$$u_j = \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & & \vdots \\ 0 & \dots & 0 & \dots & 1 \\ \vdots & & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix} \text{at the} \quad (j+1)^{\text{th}} \text{ row} \tag{1.8}$$

$$z = \begin{pmatrix} 0 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{pmatrix} \tag{1.9}$$

Denote $V := \text{span}\{v_1, ..., v_n, u_1, ..., u_n\}$, and define a skew-symmetric bilinear form by the relations on the basis as $w(v_i, u_i) = 1$ for all i, $w(v_i, u_j) = 0$ if $i \neq j$, and $w(v_i, v_j) = w(u_i, u_j) = 0$ in general.

Then, notice that under matrix commutator, it satisfies the following:

$$[v_i, v_j] = [u_i, u_j] = 0, \quad [v_i, u_i] = z = w(v_i, u_i)z, \quad \forall i, j \in \{1, ..., n\} \tag{1.10}$$

$$[v_i, u_j] = 0, \text{ if } \ i \neq j \tag{1.11}$$

Which, $\mathfrak{h}$ under matrix commutator not only is a Lie algebra, but one that satisfies the condition that $[v, u] = w(v, u)z$ for all $v, u \in V$ (since such relation is satisfied by the basis elements of V).

Then, taken the matrix exponential of the Lie algebra, since each v_i, u_j satisfies $v_i^2 = u_j^2 = z^2 = 0$ as matrix, then we automatically get that $\exp(tv_i) = \text{id}_{2n+2} + tv_i, \exp(su_j) =$

$\mathrm{id}_{2n+2} + su_j$, and $\exp(rz) = \mathrm{id}_{2n+2} + rz$. Which, Notice that they generate all matrices of the following form:

$$A = \begin{pmatrix} 1 & t_1 & \dots & t_n & r \\ 0 & 1 & \dots & 0 & s_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & s_n \\ 0 & 0 & \dots & 0 & 1 \end{pmatrix} \tag{1.12}$$

Which, this Lie group generated by exponentiation of $\mathfrak{h}$ (denoted as H), as a manifold is isomorphic to $\mathbb{R}^{2n+1}$, which is in fact simply-connected, so this is one of the desired Lie group.

(1) Given 3-dimensional real Lie algebra $\mathfrak{g}$ with basis x, y, z, consider $V = \mathrm{span}\{x, y\}$, then $\mathfrak{g} = V \oplus \mathbb{R}z$. On V, one can define a bilinear form $w : V \times V \to V$ by $w(x, x) = w(y, y) = 0$, $w(x, y) = -w(y, x) = 1$. Then, w is in fact a non-degenerate skew-symmetric bilinear form on V (since given any nonzero $ax + by \in V$, if $a \neq 0$, then $w(ax + by, y) = aw(x, y) = a \neq 0$; similar for $b \neq 0$, then choose x as a counterpart instead).

Also, given any $ax + by, cx + dy \in V$, one has $w(ax + by, cx + dy) = (ad - bc)w(x, y) = (ad - bc)$ Hence, we have the following on its Lie algebra structure:

$$[ax + by, cx + dy] = (ac - bd)[x, y] = w(ax + by, cx + dy)z \tag{1.13}$$

$$[ax + by, z] = 0 \tag{1.14}$$

Hence, the Heisenberg Algebra satisfies the given condition in part (2), which its Lie group G automatically satisfies the condition $\exp(tx)\exp(sy)\exp(-tx)\exp(-sy) = \exp(tsz)$ (with $x, y \in V$), hence $\exp(tx)\exp(sy) = \exp(tsz)\exp(sy)\exp(tx)$, which is the desired statement.

Now, for a concrete group for 3-dimensional Heisenberg Algebra, consider the group $H \subseteq \mathrm{GL}_3(\mathbb{R})$ collecting all matrices of the following:

$$H = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \in \mathrm{GL}_3(\mathbb{R}) \middle| a, b, c \in \mathbb{R} \right\} \tag{1.15}$$

Notice that as $\mathbb{R}$-manifold, $H \cong \mathbb{R}^3$ due to the freedom of 3 entries, and for any $a, b, c \in \mathbb{R}$ such matrix is always invertible because of the diagonals all being 1, and its uppertri-angular. So, H is simply-connected. Also, taking the geometric tangent vector by taking differentiation, if fixing only one entry of a, b, c, one deduces that the corresponding Lie algebra $\mathfrak{h}$ is spanned by $\left\{ x := \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, y := \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, z := \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\}$. Which, their matrix commutation is as follow:

$$[x, y] = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = z \tag{1.16}$$

$$[x, z] = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0 \tag{1.17}$$

$$[y,z] = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = 0 \tag{1.18}$$

Hence, Lie group H and its Lie algebra $\mathfrak{h}$ does satisfy the given condition in Part (1).

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Problem 2

Etingof Problem Set 3.18:
Let

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in \mathrm{SL}(2,\mathbb{C}) \tag{2.1}$$

(1) Show that $S = \exp(\frac{\pi}{2}(f-e))$, where $e, f \in \mathfrak{sl}(2,\mathbb{C})$ are standard basis elements.
(2) Compute Ad S in the basis e, f, h.

Solution:

(1) If consider the standard basis of $\mathfrak{sl}(2,\mathbb{C})$ as $\left\{h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\right\}$, if consider $\exp(t(f-e))$, we have the following (using the relation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$):

$$\begin{aligned} \exp(t(f-e)) &= \sum_{n=0}^{\infty} t^n \frac{\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^n}{n!} = \sum_{k=0}^{\infty} t^{2k} \frac{\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{2k}}{(2k)!} + t^{2k+1} \frac{\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{2k+1}}{(2k+1)!} \\ &= \sum_{k=0}^{\infty} t^{2k} \frac{\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}^k}{(2k)!} + t^{2k+1} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}^k \frac{\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}}{(2k+1)!} \\ &= \sum_{k=0}^{\infty} (-1)^k \frac{t^{2k}}{(2k)!} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + (-1)^k \frac{t^{2k+1}}{(2k+1)!} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} \cos(t) & 0 \\ 0 & \cos(t) \end{pmatrix} + \begin{pmatrix} 0 & -\sin(t) \\ \sin(t) & 0 \end{pmatrix} \end{aligned} \tag{2.2}$$

Hence, plugin $t = \frac{\pi}{2}$, we get that $\exp(\frac{\pi}{2}(f-e)) = \begin{pmatrix} \cos(\frac{\pi}{2}) & -\sin(\frac{\pi}{2}) \\ \sin(\frac{\pi}{2}) & \cos(\frac{\pi}{2}) \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = S$.

(2) If consider the exponentiation curve of the basis elements h, e, f, we get the following for all $t \in \mathbb{C}$ (recall that $e^2 = f^2 = 0$):

$$\exp(th) = \sum_{n=0}^{\infty} t^n \frac{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^n}{n!} = \sum_{n=0}^{\infty} \frac{\begin{pmatrix} t^n & 0 \\ 0 & (-t)^n \end{pmatrix}}{n!} = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix} \tag{2.3}$$

$$\exp(te) = \sum_{n=0}^{\infty} t^n \frac{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^n}{n!} = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad \exp(tf) = \sum_{n=0}^{\infty} t^n \frac{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}^n}{n!} = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix} \tag{2.4}$$

Then as a smooth map, Ad $S : \mathrm{SL}(2,\mathbb{C}) \rightsquigarrow \mathrm{SL}(2,\mathbb{C})$ acts on the above three matrices as follow:

$$\begin{aligned}\text{Ad } S(\exp(th)) &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & -e^{-t} \\ e^t & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} e^{-t} & 0 \\ 0 & e^t \end{pmatrix}\end{aligned} \tag{2.5}$$

$$\text{Ad } S(\exp(te)) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & t \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -t & 1 \end{pmatrix} \tag{2.6}$$

$$\text{Ad } S(\exp(tf)) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -t & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix} \tag{2.7}$$

Therefore, geometrically the differential $(\text{Ad } S)_*$ has the action on h, e, f given as follow:

$$\begin{aligned}(\text{Ad } S)_*(h) &= \frac{d}{dt}\bigg|_{t=0} \text{Ad } S(\exp(th)) = \frac{d}{dt}\bigg|_{t=0} \begin{pmatrix} e^{-t} & 0 \\ 0 & e^t \end{pmatrix} \\ &= \begin{pmatrix} -e^{-t} & 0 \\ 0 & e^t \end{pmatrix}\bigg|_{t=0} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = -h\end{aligned} \tag{2.8}$$

$$\begin{aligned}(\text{Ad } S)_*(e) &= \frac{d}{dt}\bigg|_{t=0} \text{Ad } S(\exp(te)) = \frac{d}{dt}\bigg|_{t=0} \begin{pmatrix} 1 & 0 \\ -t & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} = -f\end{aligned} \tag{2.9}$$

$$\begin{aligned}(\text{Ad } S)_*(f) &= \frac{d}{dt}\bigg|_{t=0} \text{Ad } S(\exp(tf)) = \frac{d}{dt}\bigg|_{t=0} \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} = -e\end{aligned} \tag{2.10}$$

So, writing in the ordered basis of e, f, h, the matrix is written as follow:

$$\mathcal{M}((\text{Ad } S)_*) = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \tag{2.11}$$

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Problem 3

Etingof Problem Set 3.19:

Let G be a complex connected Lie group.

(1) Show that $g \mapsto \mathrm{Ad}\ g$ is an analytic map $G \to \mathfrak{gl}(\mathfrak{g})$.

(2) Assume that G is compact, show that then $\mathrm{Ad}\ g = 1$ for any $g \in G$.

(3) Show that any connected compact complex group must be commutative.

(4) Show that if G is a connected complex compact group, then the exponential map gives an isomorphism of Lie groups $\mathfrak{g}/L \cong G$ for some Lattice $L \subset \mathfrak{g}$ (i.e. a free abelian group of rank equal to $2\dim(\mathfrak{g})$).

Solution:

(1) In general, it is true that $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g}) \subset \mathfrak{gl}(\mathfrak{g})$ is a Lie group homomorphism (as real Lie group, verifiedin the previous HW). Which, the reason why given G as a complex connected Lie group, such map is complex analytic, is because taking the differential $\mathrm{Ad}\ g$ when expressing the tangent space (togehter with some neighborhood of $1 \in G$) it turns into a complex analytic map, and varying g (which is also described by analytic map) has every entry of the differential also varies analytically, hence Ad is in fact an analytic map.

(2) Given that G is compact (also connected complex Lie group by initial assumption), then the analytic map $\mathrm{Ad} : G \to \mathfrak{gl}(\mathfrak{g})$ must either be surjective or constant (by the maximum principle of complex manifold). However, since $\mathfrak{gl}(\mathfrak{g}) \cong M_n(\mathbb{C})$ as manifold/vector space (given that $\dim_{\mathbb{C}}(\mathfrak{g}) = n$), $\mathfrak{gl}(\mathfrak{g})$ is not bounded with the natural euclidean norm, which is not compact. Then, $\mathrm{Ad} : G \to \mathfrak{gl}(\mathfrak{g})$ must not be surjective (if it's surjective, since $\mathrm{Ad}(G) = \mathfrak{gl}(\mathfrak{g})$, then $\mathrm{Ad}(G)$ is not compact; however Ad as a continuous map must send compact sets to compact sets, so $\mathrm{Ad}(G)$ is also compact. This is a contradiction). Hence, Ad must be constant.

Finally, since $\mathrm{id}_{\mathfrak{g}} \in \mathrm{im}(G)$ (since $1 \in G$ satisfies $\mathrm{Ad}(1) = \mathrm{id}_G$, so its differential $\mathrm{Ad}\ 1_* = \mathrm{id}_{\mathfrak{g}}$), then with Ad being constant, then one must have $\mathrm{Ad}\ g = \mathrm{id}_{\mathfrak{g}}$ for all $g \in G$. Denote $1 := \mathrm{id}_{\mathfrak{g}} \in \mathfrak{gl}(\mathfrak{g})$, one yields $\mathrm{Ad}\ g = 1$ for all $g \in G$.

(3) Based on part (2), every $g \in G$ satisfies $\mathrm{Ad}\ g = 1 \in \mathfrak{gl}(\mathfrak{g})$. However, recall that every Lie group homomorphism $\varphi : G \to K$ is uniquely determined by $\varphi_* : \mathfrak{g} \to \mathfrak{k}$ (where the mathfrak of the lowercase symbol represents the lie algebra of the corresponding Lie group). Then, with every $g \in G$ (and the corresponding adjoint map $\mathrm{Ad}\ g : G \to G$) satisfies $\mathrm{Ad}\ g_* = 1 \in \mathfrak{gl}(\mathfrak{g})$, while id_G also satisfies $(\mathrm{id}_G)_* = 1 \in \mathfrak{gl}(\mathfrak{g})$, by uniqueness of the Lie group homomorphism, $\mathrm{Ad}\ g_* = (\mathrm{id}_G)_*$ implies $\mathrm{Ad}\ g = \mathrm{id}_G$. Hence, $\mathrm{Ad}\ g(h) = ghg^{-1} = h$ for all $g, h \in G$, showing that $gh = hg$ for all $g, h \in G$, or G is commutative.

(4) Given that G is a connected complex compact Lie group, then by part (3) we know G is commutative. Hence, given any $x, y \in \mathfrak{g}$ (its Lie algebra), one has $\exp(x+y) = \exp(x)\exp(y)$ based on commutativity. Hence, $\exp : \mathfrak{g} \to G$ is in fact a group homomorphism.

Then, recall that there exists open neighborhood $U \subseteq \mathfrak{g}$ (where $0 \in U$) and $\tilde{U} \subseteq G$ (where $1 \in \tilde{U}$), such that $\exp : U \xrightarrow{\sim} \tilde{U}$ is a diffeomorphism, while open neighborhood of $1 \in G$ generates G based on the connectedness of G. Hence, every $g \in G$ can be expressed as products of finite $g_1, ..., g_n \in \tilde{U}$, where each $g_i \in \tilde{U}$ can be expressed as $\exp(x_i)$ for some $x_i \in U$. Hence, we get that $g = \prod_{i=1}^n g_i = \prod_{i=1}^n \exp(x_i) = \exp\left(\sum_{i=1}^n x_i\right) \in \mathrm{im}(\exp)$, which shows that exp is surjective.

Hence, by First Isomorphism Theorem of groups, one must have $\mathfrak{g}/\ker(\exp) \cong G$. Let $L := \ker(\exp)$, we' first claim that L is discrete. However, this simply follows from the fact that exp is a local diffeomorphism near $0 \in \mathfrak{g}$, which there is a neighborhood $U \subseteq \mathfrak{g}$ where exp is diffeomorphism. Hence, within this neighborhood it intersects L only at 0. Now, for any $v \in L$, $v+U$ is a suitable open neighborhood of $v+U$ where exp is again a diffeomorphism (translation of the original thing, and the fact that exp is a group homomorphism), hence only intersecting L at v. So, L is discrete, hence $\mathfrak{g}/L$ in fact has a complex manifold structure, that is diffeomorphic to G.

Now, we claim that L is in fact a lattice. First, it's a free abelian group, since for all $v \in L$, any $n \in \mathbb{Z}$ satisfies $n \cdot v \in L$ by the property of kernel, and the fact that it's a subgroup inside of a vector space (which is again a free abelian group).

Then, to say that L has rank $\geq 2\dim(\mathfrak{g})$, we'll prove by contradiction: Suppose L has rank $k < 2\dim(\mathfrak{g})$, then notice that when viewing $\mathfrak{g}$ as $\mathbb{R}^{2\dim(\mathfrak{g})}$ (where $\dim(\mathfrak{g})$ is the complex dimension), the list of basis $v_1, ..., v_k$ of L as a free $\mathbb{Z}$-module is not spanning $\mathfrak{g}$ as $\mathbb{R}$-vector space (because the list has length $<$ real dimension of $\mathfrak{g}$). Hence, there exists $w \in \mathfrak{g}$ such that $w \notin \operatorname{span}\{v_1, ..., v_k\}$, and $L \subset \operatorname{span}\{v_1, ..., v_k\}$ by definition. So, since exponential map quotients any line $\mathbb{R}v_1$ into S^1 (since there are discrete and evenly-spaced point on $\mathbb{R}v_1$ that gets send to the same point by the property that $v_1 \in L$, so $\mathbb{Z}v_1$ are all send to the point), then in particular as a manifold $\exp(\mathfrak{g}) \cong \mathbb{R}^{2\dim(\mathfrak{g})-k} \times (S^1)^k$ (since each basis element $v_i \in L$ has the line $\mathbb{R}v_i$ quotients to S^1 without affecting the structure of the other elements). Yet, in this case $\exp(\mathfrak{g}) = G$ is not compact (using the fact that exp is surjective in this case), which violates our assumption that G is compact. So, rank of L being $k < 2\dim(\mathfrak{g})$ reaches a contradiction.

Finally, to say that rank of L has $k \leq 2\dim(\mathfrak{g})$, suppose the contrary again that $k > 2\dim(\mathfrak{g})$. Then, one can choose a basis $v_1, ..., v_{k-1}, w$ for L, which they're $\mathbb{Z}$-linearly independent, and product a discrete set in $\mathfrak{g}$. However, since this list has length $k > 2\dim(\mathfrak{g})$ (real dimension of $\mathfrak{g}$), then it is $\mathbb{R}$-linearly dependent, hence there exists some index $1 \leq i \leq k$ and some real numbers $a_1, ..., a_i$ (not all 0), such that $a_1v_1 + ... + a_iv_i = w$.

Notice that if $a_1, ..., a_i$ are all rational numbers, then $v_1, ..., v_i, w$ are $\mathbb{Z}$-linearly dependent (since let $q = \operatorname{lcm}$ of the denominators of $a_1, ..., a_i$, then one has $q(a_1v_1 + ... + a_iv_i) = qw$, while every coefficient $qa_j \in \mathbb{Z}$ and $q \in \mathbb{Z}$ and not all of them are 0), which contradicts the assumption that they're $\mathbb{Z}$-linearly independent. Hence, at least one of them needs to be irrational.

Which, since by scaling with suitable rational numbers (i.e. lcm of denominators of all ratioanl coefficients), can assume $a_1, ..., a_i$ only contains integers and irrational numbers (also, up to shifting by integer multiples of $v_1, ..., v_i$ one can say all integer coefficients are 0). Now, for any a_j that is irrational, notice that by taking the cost $a_j + \mathbb{Z}$ and take the quatient of $\mathbb{R}/\mathbb{Z} \cong S^1$, $a_j + \mathbb{Z}$ has a dense image in S^1. Hence, if viewing S^1 as $[0,1)$ under addition quotient by $\mathbb{Z}$, every $\varepsilon > 0$ has one $n \in \mathbb{Z}$, such that $k = (n + a_j \bmod \mathbb{Z}) \in [0,1)$ satisfies $|k - 0| < \varepsilon$. Which, there exists $n' \in \mathbb{Z}$ such that $|n' + a_j - 0| < \varepsilon$ in $\mathbb{R}$ (by moving $n + a_j$ into the interval $[0,1)$ through adding some integers). Now, for every $\varepsilon > 0$, since $\|v_j\| > 0$, then $\frac{\varepsilon}{k}\|v_j\|) > 0$, hence by the statement above there exists $n_j \in \mathbb{Z}$, such that $\|n_jv_j + a_jv_j - 0\| = |n_j + a_j - 0| \cdot \|v_j\| < \|v_j\| \cdot \frac{\varepsilon}{k \cdot \|v_j\|} = \frac{\varepsilon}{k}$.

Then, given $a_{i_1}v_{i_1} + ... + a_{i_l}v_{i_l} = w$ has all coefficients being irrational, then the following is satisfied:

$$\left\| \sum_{j=1}^{l} \left(n_{i_j} + a_{i_j}\right) v_{i_j} \right\| \leq \sum_{j=1}^{l} |n_{i_j} + a_{i_j}| \cdot \|v_{i_j}\| < \sum_{j=1}^{l} \frac{\varepsilon}{k} < \varepsilon \tag{3.1}$$

(Note: this is because $l < k$ by the fact that $v_{i_1}, ..., v_{i_l}$ is a sublist of $v_1, ..., v_k$).

Notice that this shows that for every $\varepsilon > 0$, there exists such expression $\sum_{j=1}^{l}\left(n_{i_j} + a_{i_j}\right)v_{i_j} \in L$ (since initially $\sum a_iv_i \in L$, hence integer combination of v_i added to this vector is still in L), such that it is distance $< \varepsilon$ away from 0, while not 0 (since $w = \sum a_{i_j}v_{i_j}$ is $\mathbb{Z}$-linearly independent from $v_{i_1}, ..., v_{i_j}$). Hence, this shows that one can get arbitrarily close things to $0 \in L$ using elements in L, then L cannot be discrete, but this is agani a contradiction.

So, L has rank $\leq 2\dim(\mathfrak{g})$, together with the previous fact that its rank $\geq 2\dim(\mathfrak{g})$, we must have L being rank $2\dim(\mathfrak{g})$, which L is a lattice, and this finishes the proof.